Biker-Driven Navigation:
A Smarter Navigation System Built by and for Cyclists

Project Concept

1. Project Overview

This project proposes the development of a bicycle-focused navigation application continuously improved by cyclists themselves. Unlike traditional navigation applications, the proposed system would take into account factors that are particularly important to bicyclists, including bicycle type, local access rules, time-based restrictions, permits, and bicycle-accessible paths that may not be properly represented in existing navigation databases.

The goal is to create a navigation system that does not simply tell cyclists where they can go according to a generalized map, but instead helps identify where cyclists can actually and efficiently travel.

2. Problem Statement

Current navigation applications can create significant problems for cyclists because their routing systems do not always account for the realities of bicycle travel.

Applications such as Google Maps and other transportation platforms may overlook existing bicycle-accessible routes or fail to recognize restrictions and policies that apply only at certain times. As a result, cyclists may be directed onto unnecessary detours that significantly increase their travel time. This problem is especially frustrating because the route may physically exist and be usable by cyclists, yet the navigation system does not recognize it as a viable option.

Problem in one sentence:

> *“Existing navigation systems are designed around generalized mapping data rather than the specific needs, experiences, and realities of individual cyclists.”*

3. Proposed Solution

The proposed solution is a community-driven bicycle navigation application built by and for cyclists.

Rather than relying exclusively on a centralized map database, the application would allow cyclists to contribute information about routes they actually use. The system would collect and evaluate this information over time, allowing frequently used but unmapped or underrepresented paths to be considered for inclusion in the official navigation network.

- Traditional digital mapping data
- User-contributed bicycle routes
- Bicycle-specific information

- Local access rules
- Time-based restrictions
- Bicycle type and capabilities
- Community verification
- Route usage data

This approach would create a navigation system that continuously improves based on real-world cycling activity.

4. User Bicycle Profile

When registering, users would create a bicycle profile. This information would help the application determine which routes are appropriate for each cyclist.

A complete profile would be important because not every route is appropriate for every type of bicycle. For example, a route that is practical for a standard bicycle may have different restrictions or practical considerations for an electric bicycle, cargo bicycle, or another specialized bicycle.

- Type of bicycle
- Electric or non-electric bicycle
- Bicycle dimensions, where relevant
- Maximum range for electric bicycles
- Ability to use certain types of bicycle infrastructure
- Other characteristics that may affect route accessibility

5. Community-Verified Routes

One of the project's central features would be a community-based route verification system.

Cyclists would be able to travel along paths that are not currently recognized by the application's official map. When multiple users successfully travel through the same path, the system would collect this information.

Instead of allowing a single user to immediately change the official map, the application would establish a verification threshold.

> ***Unrecognized Path*** → ***Cyclists Use Path*** → ***Usage Data Collected*** → ***Community Verification*** → ***Path Reviewed*** → ***Path Added to Navigation Network***

This creates a balance between allowing the community to identify missing routes and preventing incorrect or unsafe routes from being automatically added.

At the end of a trip, users could receive information such as:

“This route has 37 verified trips. 13 more trips are needed before it can be reviewed for official inclusion.”

This would allow users to see how their participation contributes to improving the cycling network.

6. Route and Rule Awareness

A major difference between this application and conventional navigation systems would be its focus on local cycling rules and conditions.

This information could allow the navigation system to make more informed routing decisions. For example, instead of simply determining that a path exists, the system could determine: “This path exists, is accessible to this user's bicycle, and is available at the current time.”

- Time-specific access restrictions
- Paths that are open only during certain hours
- Bicycle-specific access rules
- Permit requirements
- Seasonal restrictions
- Bicycle infrastructure
- Paths that allow certain bicycle types but restrict others
- Temporary closures
- Community-reported route conditions

7. How the System Would Work

Step 1: Create a Profile — The cyclist registers and provides information about their bicycle.

Step 2: Enter a Destination — The user enters their starting point and destination.

Step 3 (Optional): Select Navigation Preferences — The user can select preferences such as avoiding ferries or major roads, preferring protected bicycle lanes, choosing shorter or faster routes, avoiding steep hills, or using bicycle-accessible paths.

Step 4: Generate a Bicycle-Specific Route — The application generates a route using both existing map information and verified community data.

Step 5: Record the Trip — With the user's permission, the application records the route traveled.

Step 6: Identify Missing Routes — If the cyclist successfully travels along a path that is not officially recognized, the system records the path.

Step 7: Build Community Evidence — Additional cyclists who use the same path contribute additional verification.

Step 8: Review the Path — Once the path reaches the required verification threshold, it can be reviewed for official inclusion.

Step 9: Update the Navigation Network — Approved paths become part of the application's official routing system.

8. Existing Alternatives

There are already several applications that provide transportation and bicycle navigation services, including Google Maps, BikeMap, and Transit.

However, these platforms may have limitations for the specific problem this project is attempting to address.

The proposed application would not necessarily attempt to replace these services immediately. Instead, it would focus on solving a specific problem: making bicycle navigation more responsive to the real-world routes and experiences of cyclists.

- Missing or outdated bicycle routes
- Delays in incorporating changes to mapping databases
- Limited bicycle-specific information
- Difficulty accounting for highly localized access rules
- Different applications relying on similar underlying mapping infrastructure
- Limited mechanisms for ordinary cyclists to help verify missing routes

9. Data Verification and Safety

Because user-generated information can be inaccurate, the project would need a verification system rather than automatically accepting every submitted route.

This would help prevent the application from recommending routes that are unsafe, illegal, inaccessible, or otherwise inappropriate.

1. Number of cyclists who successfully traveled the path
2. Frequency of use
3. Consistency of GPS data
4. Whether users successfully completed the route
5. Applicable city, park, bridge, or transportation rules
6. Temporary or permanent restrictions
7. Bicycle-type compatibility
8. Administrative or human review when necessary

10. Long-Term Vision

The long-term goal is to create a living bicycle navigation network that improves as more cyclists use it.

Instead of waiting for a centralized mapping company to discover and update every route, cyclists could actively contribute to improving the system.

The result would be a navigation platform that reflects not only the physical geography of New York City, but also the actual experience of navigating the city by bicycle.

- Bicycle routes
- Cycling infrastructure
- Route restrictions
- Time-based access rules
- Bicycle-type restrictions
- Frequently used unofficial routes
- Temporary closures
- Community-verified paths
- Route quality information

11. Project Goal

The central goal of this project is to make bicycle navigation more accurate, flexible, community-driven, and responsive to real-world cycling conditions.

The application would address a common frustration among cyclists: knowing that a usable route exists but being unable to access it through conventional navigation software.

By combining bicycle profiles, real-world route data, community verification, and local rules, the project aims to create a navigation system where the people who actually use the bicycle network can help improve the map itself.

Proposed Project Tagline

“The map for cyclists, built by cyclists.”